

EgoScore — A Curation Engine for EgoVerse

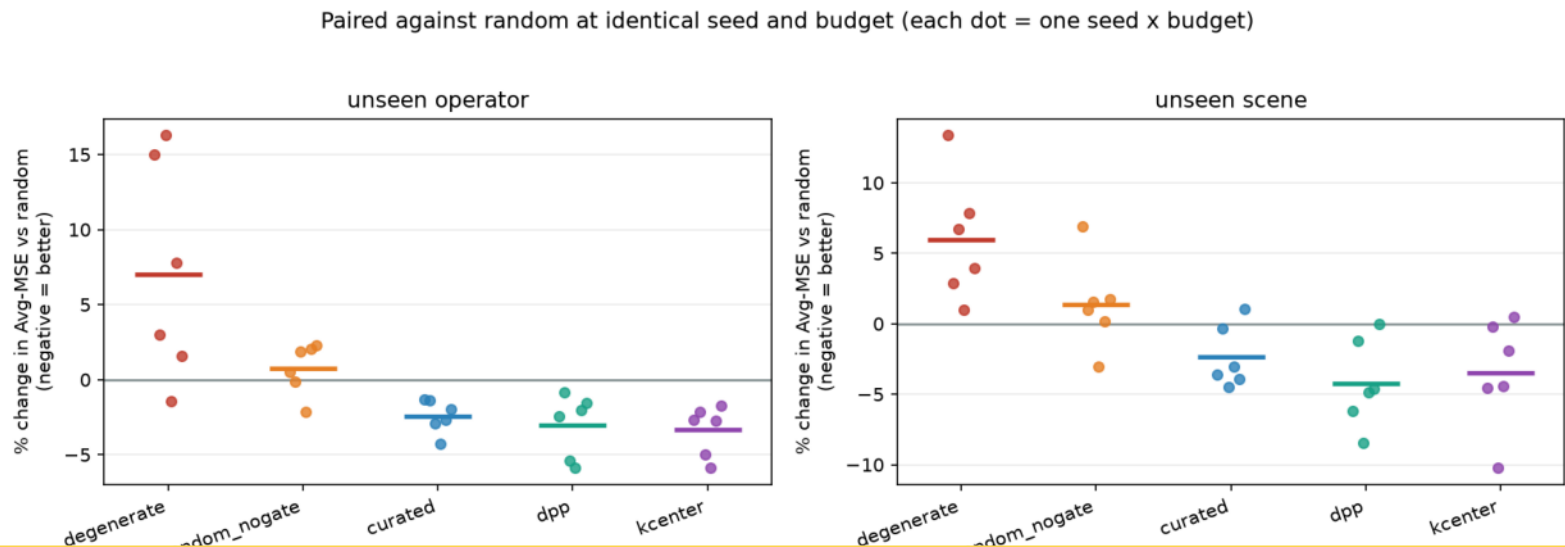
Track 1 · Which episodes are worth training on? · github.com/sahilmahendrakar/egoscore

Claim tested: at a fixed budget K , a quality-gated, coverage-maximizing subset trains a better policy than K random episodes.

Result

3 seeds × 2 budgets × 2 held-out axes = 12 paired tests

SELECTOR	WINS	AVG-MSE
dpp (log-det diversity)	12/12	-3.63%
kcenter	11/12	-3.42%
curated (facility location)	11/12	-2.40%
no quality gate	3/12	+1.05%
degenerate (control)	1/12	+6.50%



Not “throw away 75% of your data for free” — the full gated pool still wins.
At $\frac{1}{4}$ budget, diversity selection closes 43% (unseen operator) and 59% (unseen scene) of the gap between a random quarter and using everything.

Why the control matters

A 3% margin on a proxy metric is easy to disbelieve. So we added a condition that should lose: the same budget concentrated into ~ 10 operator $\times$ scene groups instead of ~ 60 .

At $K=25\%$ it is +13.0% (unseen operator) / +9.3% (unseen scene) — large, unambiguous, correctly signed. The harness detects the diversity effect the EgoVerse paper reported, so the smaller margin is a measurement, not noise.

Three things we found in the data

Zarr arrays are zero-padded to a chunk boundary. Read them raw and the trailing frames look exactly like a frozen tracker — the gate would have discarded the entire dataset. True length is `total_frames` in `attrs`.

Images cost 84.6 GB for this slice; poses cost 2.3 GB. Every signal here is pose-only, deliberately: curation must cost less than the training it saves.

An “episode” is not a common unit: median 93 s in `rl2` vs 6.6 s in `mecka`. Episode-count budgets do not transfer across labs.

Limitations

Avg-MSE is a proxy, not robot success — the EgoVerse authors say so themselves, and we adopt it on those terms.

The proxy policy is proprioceptive, not visual. One task, one lab — no claim of transfer.

Facility location was our a priori pick and it lost to DPP and k-center. Reported, not quietly reordered.

Method: `rl2 fold_clothes` — 572 episodes, 20 operators $\times$ 16 scenes, the only EgoVerse slice with a populated operator $\times$ scene grid (microagi has no operator/scene metadata; mecka has 2 scenes).
Proxy: offline Avg-MSE (the EgoVerse authors' own metric) from a ridge action-chunk policy — 10-frame proprio history $\rightarrow$ 30-step future EE pose, relative to current pose.